Real-time tumor tracking for MR-guided radiotherapy using Segment Anything Model 2

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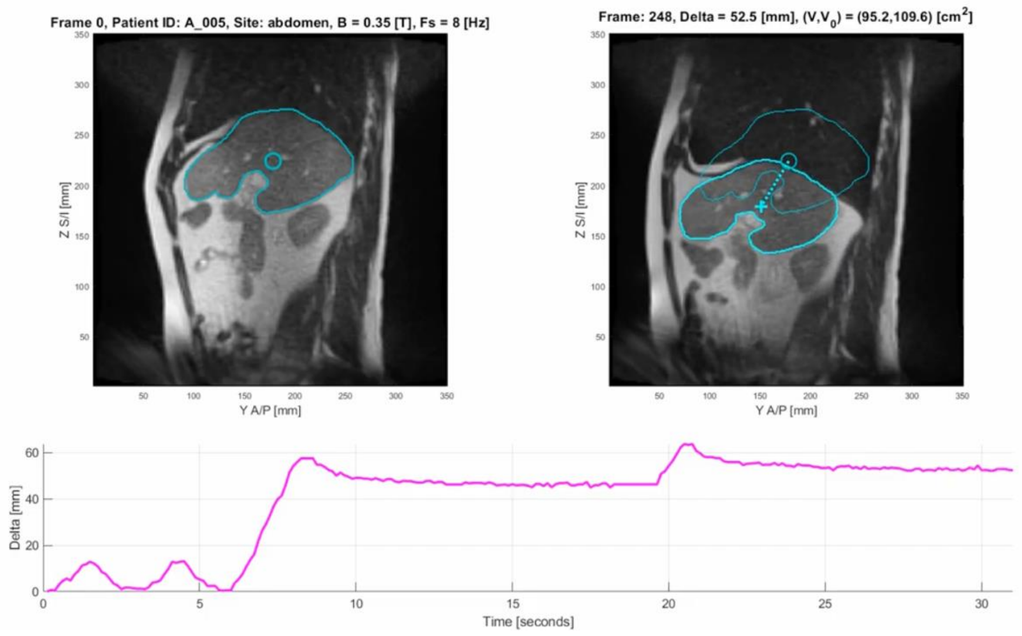
INTRODUCTION

Magnetic resonance-guided radiotherapy (MRgRT) is an emerging field of cancer treatment. Hybrid machines that combine MRI with a linear accelerator (MR-linacs) can visualize soft tissue at frame rates of 1 to 8 Hz [1]. Accurate tracking of tumors that move and deform throughout motion (e.g., during respiratory cycles) enables precise delivery of higher radiation doses while sparing healthy tissue.

Existing clinical software struggles with large non-rigid motion. Standard practice is to use a gating procedure, turning off the beam when large motion is detected. This reduces the duty cycle of the beam and extends treatment times. Deep learning models are promising for this task due to their robust performance and fast inference time.

Segment Anything Model 2 (SAM2) is a foundation model for object segmentation in images and videos [2]. Given an image and prompt, SAM2 predicts a segmentation mask. The prompt can be a bounding box around an object of interest, or a segmentation mask from another frame in a video. Taking previous frames and masks as input allows SAM2 to track moving objects by predicting segmentations frame-by-frame.

FIGURE 1: TUMOR MOTION AND DEFORMATION CAPTURED BY TIME-SERIES 2D MRI



OBJECTIVES

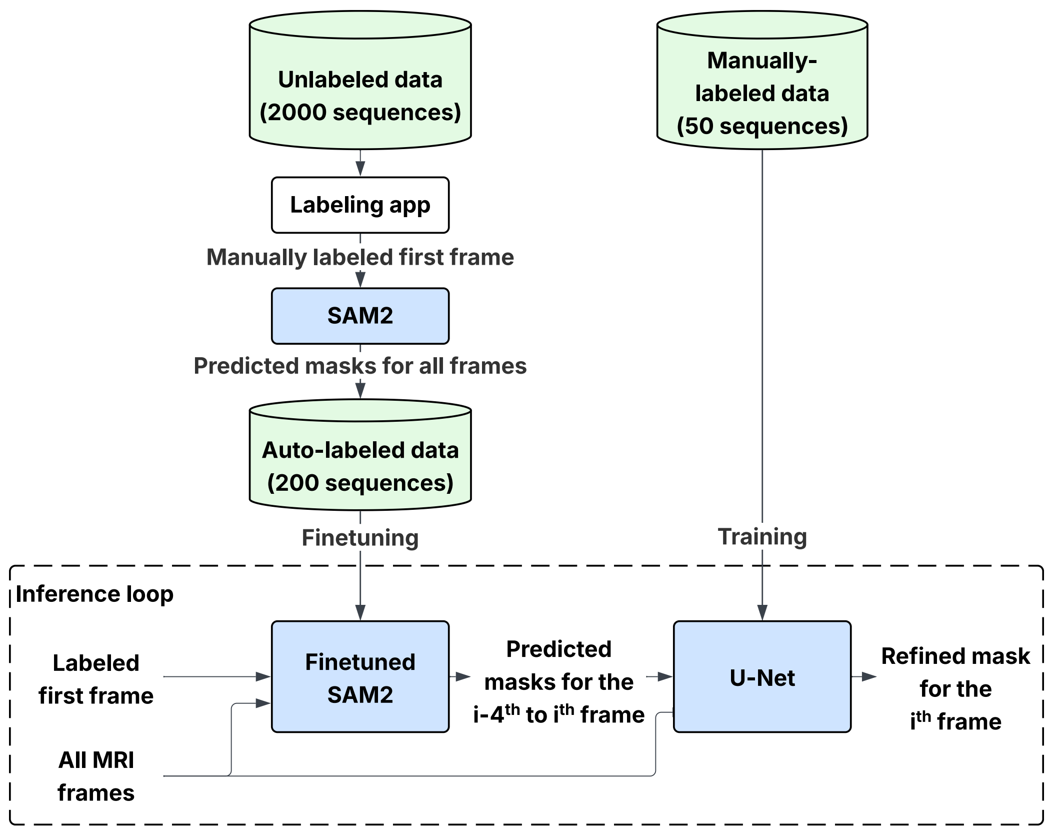
- Track tumors across 2D sagittal cine-MRI sequences, given a segmentation mask for the first frame
- The method must be robust to different frame rates (1 to 8 Hz), field strengths (1.5 and 3T), and anatomical regions (chest, abdomen, etc.)
- Runtime must be under 1 second/frame on an Nvidia A10G GPU (representative hardware for a clinical imaging workstation) to enable real-time use

METHODS

Models were trained using TrackRAD2025, an open dataset of 2D sagittal cine-MRI (time-series) sequences from MR-linacs across multiple institutions [3]. The dataset includes 50 labeled sequences from 50 patients, totaling 5027 frames, and over 2000 unlabeled sequences from 477 patients, totaling over 2.8 million frames.

In the inference loop, SAM2 was given the MRI sequence and a segmentation mask of the first frame to predict masks for all subsequent frames. Predictions were refined using a U-Net with a sliding window approach. The image and predicted mask for the current and previous four frames were encoded as a 10-channel input to predict the true mask. The U-Net was trained on the 50 labeled sequences.

FIGURE 2: MODEL ARCHITECTURE



SAM2 was also leveraged to rapidly annotate images to finetune itself. A web application was developed using Gradio, a popular user interface library [4] and deployed using HuggingFace, a cloud hosting service for machine learning applications. The app displayed an MRI sequence and allowed users to draw bounding boxes on the first frame. SAM2 then predicted a segmentation within the box, which annotators could iteratively refine. When satisfied, users submitted the final segmentation, and SAM2 predicted masks across the sequence. Annotators did not have advanced training in reading MRI, and the annotations were not validated. Using this approach, 200 of the over 2000 unlabeled sequences totaling 94,441 frames were labeled.

Training was done on the Mayo Clinic Radiation Oncology HPC cluster. The model was evaluated on image similarity metrics and runtime.

RESULTS

TABLE 1: SEGMENTATION PERFORMANCE

Model	Dice	P95 Hausdorff distance (pixels)	Euclidean center distance (pixels)	Runtime per frame (s)
SAM2 zero-shot	0.9158	3.324	1.390	0.0199
Finetuned	0.9195	3.187	1.306	0.0199
Finetuned + U-Net	0.9197	3.066	1.258	0.164

FIGURE 3: PERCENT PERFORMANCE IMPROVEMENT RELATIVE TO ZERO-SHOT SAM2

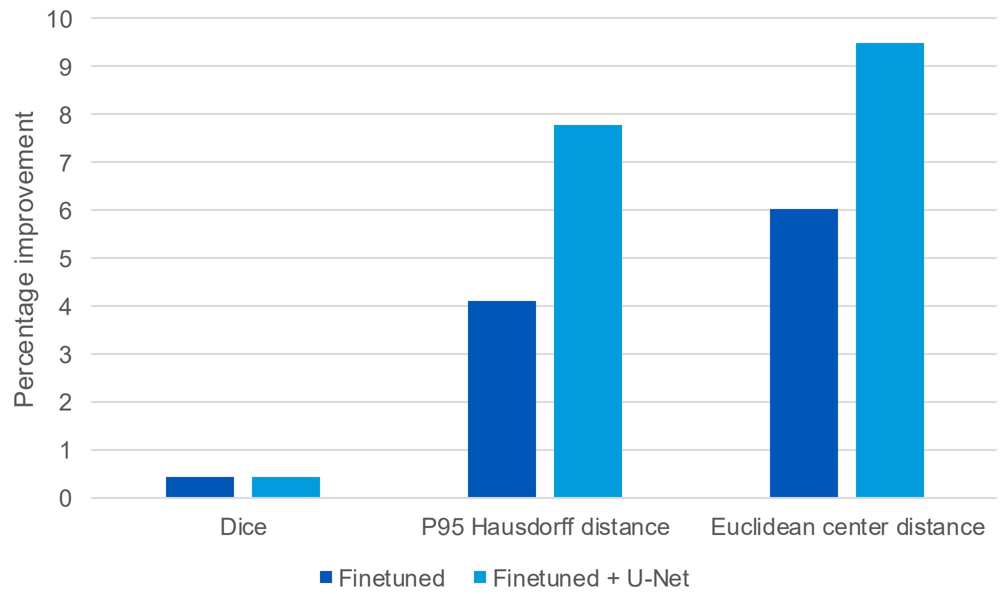
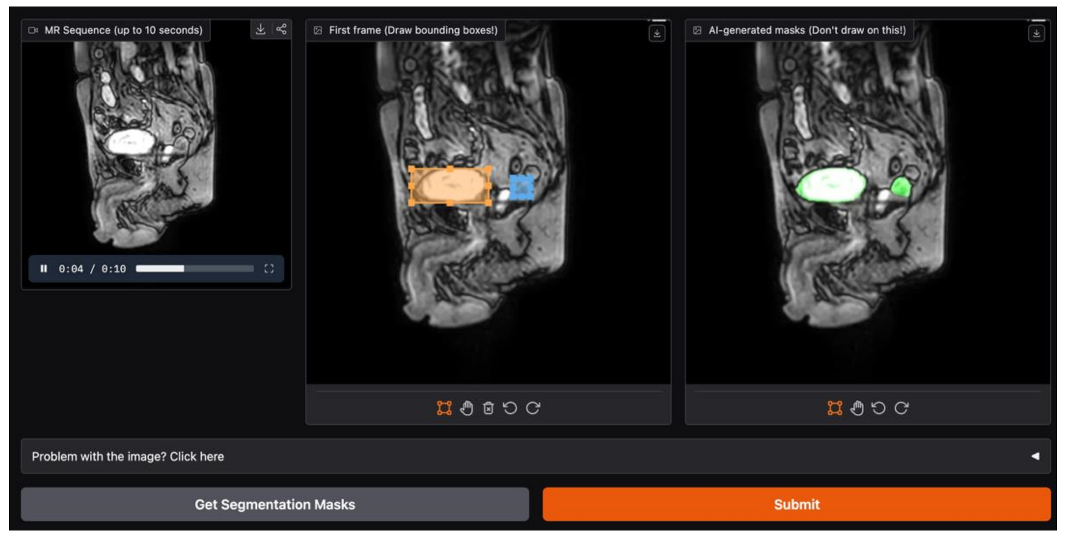


FIGURE 4: IMAGE ANNOTATION WEB APPLICATION



User interface with (left) the cine-MRI sequence as an interactive video, (center) user-defined bounding boxes on objects of interest, and (right) masks predicted by SAM2. The application is hosted on HuggingFace Spaces, and input and output data was stored in HuggingFace Datasets. SAM2 inference runs on a cloud GPU.

DISCUSSION

- Training data quality was a major limitation. Annotators were not imaging specialists, and no validation was done
- Automated rules-based validation (e.g., to identify frames with disappearing objects, unrealistic motion) would improve data quality
- Validation could be extended to an active learning pipeline, where bad frames are sent to the annotation tool for relabeling
- Leveraging the active open-source ecosystem around Gradio and HuggingFace enabled development of the annotation tool with a quick turnaround (2 weeks)
- Implementing the image annotation tool as a web application powered by cloud GPUs made it usable on any computer with an internet connection, without requiring software or hardware installation

CONCLUSIONS

- SAM2 accurately tracks objects in 2D cine-MRI in real-time
- SAM2's strong zero-shot performance was improved through fine-tuning and mask refinement
- A sliding window approach employing multiple channels can encode motion information for a U-Net
- Frameworks such as Gradio and HuggingFace enable rapid development and deployment of custom medical imaging software applications

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